

ANTIBACTERIAL DRUGS

(Anti-mycobacterials)

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OVERVIEW

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- ▶ Mycobacteria are acid fast bacilli that cause a variety of diseases including
 1. Tuberculosis (TB)
 2. Leprosy
 3. *Mycobacterium avium* intracellular infections (localized or disseminated)
- ▶ T.B is caused by *Mycobacterium tuberculosis* of which about one third of the world population is latently infected with T.B courtesy of the competent host immunity
- ▶ This latent infection is likely to develop into an active infection in 10% lifetime of individuals having an intact immunity
- ▶ *Mycobacterium avium* infections are most frequently seen in immunocompromised patients e.g. HIV/AIDS patients mostly presenting as a form of pulmonary disease, lymphadenitis or bacteremia
- ▶ Leprosy or Hansen's disease is caused by *Mycobacterium leprae* which infects the skin and peripheral nervous system

Mode of Transmission

- ▶ The respiratory tract is the principal route of infection through inhalation of infected air droplet nuclei

CLASSIFICATION OF TB

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Classification of T.B is done based on the following;

- i. Bacteriological status
 - Bacteriologically confirmed
 - Clinically diagnosed
- ii. Classification based on previous TB treatment – Relapse, failure, loss to follow up, other previously treated patients
- iii. Classification based on HIV status – HIV TB patient, HIV negative TB patient, HIV status unknown TB patient
- iv. Classification based on drug resistance – Mono resistant, Rifampicin resistant, Isoniazid resistant, Poly drug resistant, Multi drug resistant (MDR), Pre-XDR, XDR
- v. Classification based on anatomical site of disease
 - Pulmonary T.B; - localised within the airway
 - Extrapulmonary T.B; - affecting other parts away from the airway

Classical symptoms of T.B

- ▶ Prolonged coughing of more than 2 weeks which is productive in nature with or without hemoptysis
- ▶ Fever
- ▶ Drenching night sweats
- ▶ Loss of appetite
- ▶ Loss of weight
- ▶ Fatigue

ANTI – TB DRUGS

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- ▶ In terms of treatment or sensitivity of the mycobacteria to drugs, T.B is further classified as either susceptible TB or Multi drug resistant (MDR) T.B
- ▶ Based on the category of T.B being treated, anti – TB drugs are referred to as either first or second line drugs
- ▶ Multi drug therapy for a period of at least 6 months is required to eradicate the pathogen and also prevent the emergence of resistance by the mycobacterium and thus monotherapy is not promoted
- ▶ Compliance and adherence to treatment are key and because of this direct observed therapy (DOTS) has been found effective where a health care provider observes each drug administered

Goals of TB treatment

Treatment of TB is focused as achieving both the public health and the therapeutic desired outcome and to achieve this below is the treatment goal of TB;

- ▶ To kill tubercle bacilli rapidly leading to elimination of persistent bacilli
- ▶ Prevent relapse of TB infection
- ▶ Prevent disease transmission
- ▶ Enhance adherence to TB treatment course

First line drugs

These are the initially used drugs to treat most TB patients

- ▶ Isoniazid
- ▶ Ethambutol
- ▶ Rifampicin
- ▶ Pyrazinamide

Second line drugs

These are reserved to treat patients infected with organisms that are resistant to first line drugs

- ▶ Fluoroquinolones – levofloxacin, moxifloxacin
- ▶ Cycloserine
- ▶ Ethionamide
- ▶ Aminosalicic acid
- ▶ Isoniazid (High dose)
- ▶ Meropenem
- ▶ Imipenem-cilastatin
- ▶ Aminoglycosides – Kanamycin, amikacin, capreomycin
- ▶ Delamanid
- ▶ Bedaqualine
- ▶ Linezolid
- ▶ etc

ISONIAZID

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- ▶ Isoniazid is a nicotinic acid derivative also called isonicotinic acid hydrazine (INH) and it has remained the mainstay of most TB regimens
- ▶ Isoniazid is widely distributed to all tissues with sufficient concentrations likely to be effective against organisms inside cells and caseous
- ▶ It is extensively metabolized with both the parent drug and its metabolized being excreted in urine
- ▶ The primary metabolite acetyl isoniazid is formed by conjugation of acetate with isoniazid in a process catalyzed by the enzyme called acetyltransferase
- ▶ The enzyme acetyl transferase is genetically determined where slow acetylators tend to have increased plasma concentration compared to the fast acetylators
- ▶ Small amount of isoniazid is converted to isonicotinic acid and acetylhydrazine
- ▶ Acetylhydrazine is believed to be the causer of hepatic toxicity and peripheral neuropathy

Mechanism of action of Isoniazid

Isoniazid inhibits the synthesis of mycolic acid, a fatty acid which is component of the mycobacteria cell wall and is responsible for acid fast staining properties

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Indications

- ▶ Isoniazid is bactericidal against sensitive strains of *M.tuberculosis* and some strains of *M.kansaii*, has little activity against *M.avium intracellure* while it has no active against *M.leprae*
- ▶ Isoniazid is part of the standard four drug regimen for TB in persons without known or suspected resistance to isoniazid which are believed to kill different populations of TB bacilli
- ▶ Isoniazid together with rifampicin, pyrazinamide eradicate the rapidly growing organisms during the first two months of treatment
- ▶ In the next four months of treatment, isoniazid and rifampicin act against persistent bacteria that slowly revert to actively growing forms
- ▶ Isoniazid is also given in latent TB as prophylaxis in persons who are at high risk of contracting TB e.g. HIV patients, recently infected patients with positive tuberculin skin test in the past 2 years

Adverse effects of Isoniazid

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- ▶ Isoniazid causes drug induced hepatitis and so patients taking it should have their serum transaminase levels be regularly monitored
- ▶ Peripheral neuropathy – this usually occurs due to pyridoxine deficiencies due to its direct inactivation by isoniazid
- ▶ Administering of pyridoxine supplements in patients taking isoniazid helps in preventing of peripheral neuropathy
- ▶ In rare cases, isoniazid causes CNS effects like toxic encephalopathy or seizures
- ▶ Haematologic abnormalities like granulocytosis, anemia or thrombocytopenia can occur

ETHAMBUTOL

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- ▶ Ethambutol is a butanol derivative that has bacteriostatic activity against mycobacterial organisms
- ▶ Administered orally, undergoes hepatic biotransformation and excreted in urine and stool

Indications

- ▶ Used in combination with other drugs to treat TB or M.avium intracellulare infections

Adverse effects

- ▶ Optic neuritis
- ▶ Visual acuity which involves impaired red-green color discrimination and hence it should not be used in children who are not able to tell the changes in colour differentiation
- ▶ Hyperuricemia and gout
- ▶ Thrombocytopenia

PYRAZINAMIDE

- ▶ Pyrazinamide is nicotinamide derivative
- ▶ It is an important drug in TB therapy because of its more rapid bactericidal and sterilizing effect

Mechanism of action

- ▶ As nicotinamide derivative, pyrazinamide is converted to pyrazinoic acid by susceptible mycobacteria
- ▶ This pyrazinoic acid inhibits mycobacterium growth by inhibiting fatty acid synthesis

Indication

- ▶ Administered in combination with rifampicin, isoniazid and ethambutol in treatment of TB
- ▶ Its inclusion in the TB regimen made it possible to reduce the duration of treatment to 6 months from the earlier 9 to 12 months

Adverse effects for pyrazinamide

- ▶ Arthralgia
- ▶ Hyperuricemia and gout
- ▶ Drug induced hepatitis
- ▶ Hematologic toxicity
- ▶ Increase in serum iron concentration

RIFAMPICIN

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- ▶ Rifampicin is a derivative of the antibiotic rifamycin which has improved pharmacokinetic properties than its parent drug
- ▶ It is rapidly absorbed after oral administration and converted to desacetyl rifampin
- ▶ Rifampicin and its metabolite are widely distributed to tissues and fluids including lung tissue, saliva and peritoneal and pleural fluids

Mechanism of action

- ▶ Rifampicin binds to β subunit of DNA dependent RNA polymerase enzyme
- ▶ This prevents the enzyme from binding to DNA and therefore, inhibits DNA transcription and RNA synthesis

Indications of rifampicin

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- ▶ Rifampicin is a broad spectrum antibiotic that has significant activity against many gram negative, gram positive and acid fast bacilli like *M.tuberculosis*, *M.kansaii*, *M.avium intracellulare*, *M.leprae*
- ▶ Administered in combination with isoniazid, pyrazinamide and ethambutol to treat TB
- ▶ Given as an alternative to isoniazid in latent TB when resistance to isoniazid is known or suspected while it is given in combination with dapson or clofazimine to treat leprosy
- ▶ Given to individuals exposed to *Haemophilus influenza* type b and likely to transmit the disease to unvaccinated children aged 4 years and younger
- ▶ Prevention of meningococcal disease in individuals who would have come in contact with patients infected with *Neisseria meningitidis* though ciprofloxacin and ceftriaxone are mostly preferred
- ▶ Rifampicin penetrates inflamed meninges and hence can be used to treat tubercular meningitis
- ▶ Administered in combination with vancomycin and gentamycin in the treatment of staphylococcal endocarditis

Adverse effects

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- ▶ Drug induced hepatotoxicity – liver function tests should be done regularly and like the other hepatotoxic drugs, rifampicin should be discontinued if signs of hepatic dysfunction become evident
- ▶ Consumption of alcohol by individuals taking rifampicin appears to cause increase in liver toxicity or drug induced hepatitis
- ▶ Hypersensitive reaction – manifesting as a flu like illness with chills, fever, fatigue and headache, this reaction being common in those on intermittent treatment who take large doses 2 to 3 times a week
- ▶ Reddish-orange or reddish-brown discoloration of saliva, tears and urine and can also cause permanent staining of soft contact lenses
- ▶ Rifampicin induces cytochrome P450 isoenzymes CYP1A2, CYP2C9 and CYP3A4 and hence can accelerate the metabolism of drugs like benzodiazepines, calcium channel blockers, digoxin, oestrogen based contraceptives, theophylline, warfarin thereby reducing their serum concentration and their subsequent effectiveness

Modality of susceptible TB treatment course

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- ▶ Susceptible TB treatment is done for a period of 6 months which is divided into two phases namely;

1. Intensive phase

- ▶ This involves the administration of a combination of four drugs i.e. rifampicin, isoniazid, pyrazinamide and ethambutol
- ▶ This is done for a period of 2 months and it is meant to eradicate the rapidly growing organisms

2. Extensive phase

- ▶ This involves the administration of rifampicin and isoniazid
- ▶ It is done for the next 4 months and it is meant to act against persistent bacteria which if not attacked can slowly revert to actively growing forms

NOTE:

- ▶ *TB meningitis, tuberculoma, osteoarticular, ocular and spinal TB are treated for a period of 12 months (2 months for intensive and 10 months for extensive phases respectively)*
- ▶ *Steroids are also considered for these types of TB including, TB pericarditis, TB IRIS (Immune reconstitution inflammatory syndrome), massive pleural effusion, massive lymphadenopathy with pressure effects, severe hypersensitivity reactions to anti-Tb drugs*

MULTI DRUG RESISTANT TUBERCULOSIS (MDR TB)

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- ▶ Multi drug resistant TB is the type of TB which become resistant to rifampicin and isoniazid
- ▶ Pre XDR TB (Pre extensively drug resistant TB) is MDR TB and also rifampicin resistant that has also become resistant to any of the fluoroquinolones
- ▶ XDR TB (Extensively drug resistant TB) is MDR TB and rifampicin resistant TB that has also become resistant to any fluoroquinolones and at least one additional Group A drugs which are the most potent second line anti-TB drugs
- ▶ Previously the treatment for MDR was 18 months to 24 months but currently a shorter regimen can only go on up to 9 months
- ▶ A shorter regimen requires at least a minimum of 6 drugs
- ▶ For Pre – XDR and XDR TB, individualized treatment regimens are used according to the drugs to which the mycobacteria is still sensitive to

Drugs used in MDR TB

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Group A – Fluoroquinolones like levofloxacin, moxifloxacin, gatifloxacin and other drugs like; bedaquiline and linezolid

Group B – Clofazimine, cycloserine or Terizidone

Group C – Ethambutol
Delamanid
Pyrazinamide
Imipenem-cilstatin
Meropenem
Amikacin
Streptomycin
Ethionamide/Prothionamide
p-aminosalicylic

DRUGS USED IN MYCOBACTERIUM AVIUM- INTRACELLULARE INFECTIONS

- ▶ Drugs that are active against M.avium intracellulare (Mycobacterium avium complex [MAC]) infections are the following;
- ▶ Ethambutol, rifampicin, rifabutin, azithromycin and clarithromycin, ciprofloxacin, streptomycin and amikacin
- ▶ Azithromycin, clarithromycin or rifabutin are used alone to prevent MAC infection in HIV infected and other immunocompromised patients
- ▶ A combination of azithromycin, ethambutol and rifampicin or rifabutin is used to treat MAC with rifampicin preferred in HIV patients while rifabutin is preferred in immuno competent patients
- ▶ Clofazimine, streptomycin or amikacin is an alternative drug for treating MAC infections in immunocompromised patients when other drugs are ineffective or not tolerated

END